

4 (short)

A portion of a bamboo scaffolding structure is idealized in Fig. 4(a). The bamboo members can be assumed to be prismatic and have the cross-section dimensions shown in Fig. 4(b). The bamboo has a Young's modulus of 15 GPa. The vertical bamboo struts are each subject to a vertical load P applied concentric to the strut.

(a) Calculate the Euler buckling load of the vertical struts. [7]

(b) Why is the buckling load likely to be less than the load calculated in part a)? [3]

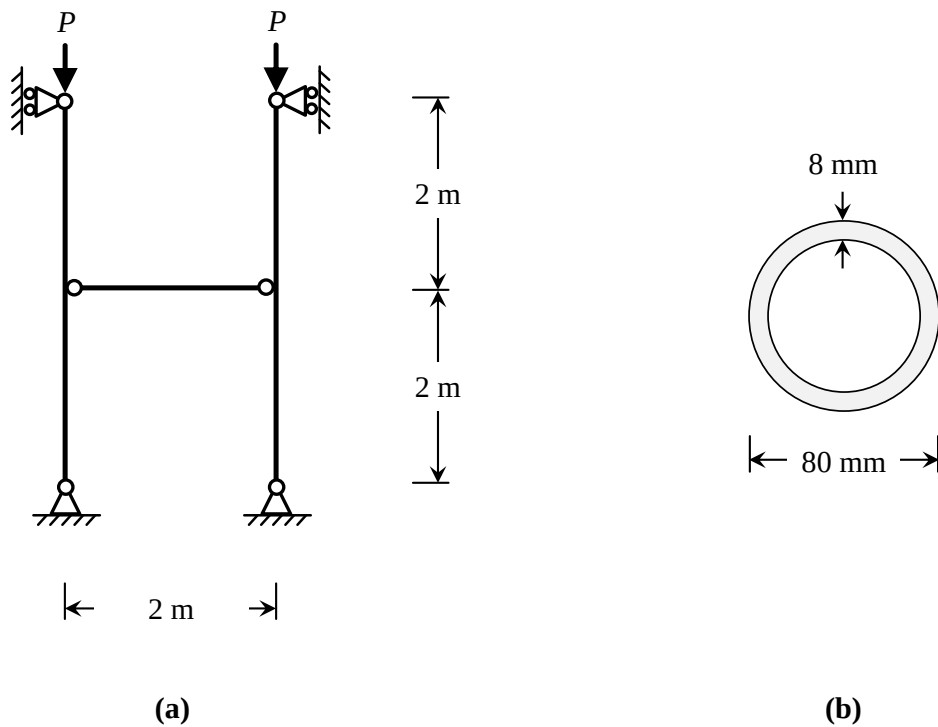


Fig. 4